

Deep Dive: 15. File Handling

Concept Summary

File handling allows you to read and write permanent data to the hard drive. Using context managers ('with') ensures safety.

Coding Structure

```
with open('filename', 'mode') as file:  
    file.write(data)
```

Example Code

```
with open('log.txt', 'w') as f:  
    f.write('System started.\n')  
  
with open('log.txt', 'r') as f:  
    print(f.read())
```

Code Explanation

- Line 1: Opens (or creates) 'log.txt' in Write ('w') mode. The 'with' keyword acts as a Context Manager—it automatically closes the file when the block ends!
- Line 2: Writes text to the file.
- Line 4: Opens the file in Read ('r') mode.
- Line 5: Reads the entire file contents and prints it.

Practice Problems

1. Write a script that appends ('a' mode) your name to a text file.
2. Read a text file and print it line-by-line using a for-loop on the file object.